

Guillermo Terrén-Serrano, PhD

Phone: +1 (505) 450-6215
Email: guille.terren@gmail.com

Webpage: gterren.github.io
GitHub: github.com/gterren

Education

University of New Mexico (UNM), Albuquerque, NM, USA. <i>Ph.D. in Engineering “pass with distinction” Electrical and Computer Engineering Department.</i> • <i>Dissertation:</i> Intra-hour Solar Forecasting using Cloud Dynamics Features Extracted from Ground-Based Infrared Sky Images.	2016 - 2022 Advisor: Prof. Martínez-Ramón
University of New Mexico, Albuquerque, NM, USA. <i>M.Sc. in Power & Energy Electrical and Computer Engineering Department.</i> • <i>Thesis:</i> Machine Learning Approach for Global Solar Radiation Time-Series Forecasting.	2015 - 2016 Advisor: Prof. Martínez-Ramón
Universidad de Zaragoza (UZ), Zaragoza, Aragón, Spain. <i>B.Sc. in Technical Industrial Engineering Escuela de Ingeniería y Arquitectura.</i> • <i>Academic Exchange:</i> Pontifícia Universidade Católica do Paraná (PUCPR), Curitiba, PR, Brazil • <i>Senior Design Project:</i> Energy-efficiency strategy for a building complex based on renewable energy, LED lighting, and passive architecture.	2006 - 2011 Advisor: Prof. Ibáñez-Carabantes

Experience

University of California, Santa Barbara (UCSB), CA, USA. <i>Climate Innovation Postdoctoral Fellow Environmental Markets Lab (emLab).</i> • <i>Research:</i> Planning reserve margin and operating reserves quantification. • <i>Description:</i> Long-term and short-term operational planning of low-carbon emissions electricity systems.	2022 - Present Supervisor: Prof. Deshmukh
University of California, Santa Barbara, CA, USA. <i>Lecturer Environmental Studies Program.</i> • <i>Teaching:</i> ENV S105 Renewable Energy Systems. Undergraduate level course.	Fall 2024 Supervisor: Prof. Deshmukh
University of California, Santa Barbara, CA, USA. <i>Postdoctoral Research Associate Statistics and Applied Probability Department.</i> • <i>Description:</i> Allocating risk to generators for price formation in electricity markets with high renewable energy participation. • <i>Research:</i> Operational Risk Financialization of Electricity Under Stochasticity (ORFEUS), Princeton University’s PERFORM team.	2022 - 2023 Supervisor: Prof. Ludkovski
University of New Mexico, Albuquerque, NM, USA. <i>Graduate Assistant Electrical and Computer Engineering Department.</i> • <i>Teaching Assistant:</i> Circuit Analysis, Electronic Circuits, Signals and Systems, and Data Structures and Algorithms. • <i>Research Assistant:</i> Machine learning for energy meteorology in smart grids at the Center for Emerging Energy Technologies (CEET).	2015 - 2022 Supervisor: Prof. Martínez-Ramón
Enerland Group, Zaragoza, Aragón, Spain. <i>Project Engineer Internship Renewable energy EPC company.</i> • <i>Projects:</i> Solar photovoltaic (PV) plants international technical and commercial project proposals (56MW, 1MW, 2.5MW, and 600kW).	2012 - 2013 Supervisor: Mr. Ballet-Sala
Institute of Technology for the Development (LACTEC), Curitiba, PR, Brazil. <i>Undergraduate Research Assistant Electronic Engineering Department.</i> • <i>Projects:</i> 30kW PV system (on-grid), and PV-powered wireless sensor for real-time measurements on transmission lines.	2012 Supervisor: Prof. Zambenedetti

Academic Publications

A Resource Adequacy Study of India’s Low Carbon Electricity System <i>Working paper</i> , 2025. A. Sharma and G. Terrén-Serrano, et al.	
Contribution of Renewable Energy Resource to the Reliability in India’s Low Carbon Electricity System. <i>Working paper</i> , 2025. G. Terrén-Serrano et al.	
Intra-Day Probabilistic Forecasting Dynamic Update for Renewable Energy Generation. <i>Working paper</i> , 2025. G. Terrén-Serrano and M. Ludkovski.	
Effects of Technology Costs and Climate Targets in India’s Transition to a Low Carbon Electricity System. <i>Working paper</i> , 2025. G. Terrén-Serrano et al.	
[TDM25] Probabilistic Day-Ahead Forecasting of System-Level Renewable Energy and Electricity Demand. <i>Nature Communications (Under review)</i> , 2025. G. Terrén-Serrano, R. Deshmukh and M. Martínez-Ramón.	
[TL25] Extreme Day-Ahead Renewables Scenario Selection in Power Grid Operations. <i>Applied Energy</i> , 2025. G. Terrén-Serrano and M. Ludkovski.	
[Pal+23] Advances in Solar Forecasting: Computer Vision with Deep Learning. <i>Advances in Applied Energy</i> , 2023, Q. Paletta, G. Terrén-Serrano, et al.	
[TM23a] Deep Learning for Intra-Hour Solar Forecasting with Fusion of Features Extracted from Infrared Sky Images. <i>Information Fusion</i> , 2023. G. Terrén-Serrano and M. Martínez-Ramón.	
[TM23b] Detection of Multiple Wind Velocity Fields using Ground-based Infrared Sky Images. <i>Knowledge-Based Systems</i> , 2023. G. Terrén-Serrano and M. Martínez-Ramón.	
[TM23c] Kernel Learning for Intra-Hour Solar Forecasting with Infrared sky images and Cloud dynamic feature extraction. <i>Renewable and Sustainable Energy Reviews</i> , 2023. G. Terrén-Serrano and M. Martínez-Ramón.	

[TM23d] **Processing of Global Solar Irradiance and Ground-Based Infrared Sky Images for Solar Nowcasting and Intra-Hour Forecasting Applications.** *Solar Energy*, 2023. G. Terrén-Serrano and M. Martínez-Ramón.

[TM22] **Geospatial Perspective Reprojections for Ground-Based Sky Imaging Systems.** *IEEE Transactions on Geoscience and Remote Sensing*, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

[TM21a] **Comparative Analysis of Methods for Cloud Segmentation in Ground-Based Infrared Images.** *Renewable Energy*, 2021. G. Terrén-Serrano and M. Martínez-Ramón.

[TM21b] **Multi-Layer Wind Velocity Field Visualization in Infrared Images of Clouds for Solar Irradiance Forecasting.** *Applied Energy*, 2021. G. Terrén-Serrano and M. Martínez-Ramón.

[Ter+21] **Girasol, a Sky Imaging and Global Solar Irradiance Dataset.** *Data in Brief*, 2021. G. Terrén-Serrano, A. Bashir, et al.

[Mam+19] **An Experimental Method to Merge Far-field Images from Multiple Long-Wave Infrared Sensors for Short-term Solar Forecasting.** *Solar Energy*, 2020. A. Mammoli, G. Terrén-Serrano, et al.

[Gar+18] **Evaluation of Dimensionality Reduction Methods Applied to Numerical Weather Models for Solar Radiation Forecasting.** *Engineering Applications of Artificial Intelligence*, 2018. O. García-Hinde, G. Terrén-Serrano, et al.

Conference Proceedings

Day-Ahead Operational Forecast of Aggregated Solar Generation Assimilating Mesoscale Meteorology Information. *IEEE PES Grid Edge Technologies Conference & Exposition*, San Diego, USA, 2025. G. Terrén-Serrano, R. Deshmukh, and M. Martínez-Ramón.

Explicit Basis Function Kernel Methods for Cloud Segmentation in Infrared Images. *Energy Reports*, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Wind Flow Estimation in Thermal Sky Images for Sun Occlusion Prediction. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Segmentation Algorithms for Ground-Based Infrared Cloud Images. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Presentations

2025 Forecasting & Markets Workshop (organized by ESIG). *Presentation*. Nashville, TN, USA.

2025 UCSB Institute for Energy Efficiency (IEE). *Seminar*. CA, USA.

2024 2024 ARPA-E Energy Innovation Summit. *Exhibition booth*. Dallas, TX, USA.

2024 Macro-Energy Systems (MES) Workshop. *Flash talk + poster*. Princeton University, NJ, USA.

2024 Massachusetts Institute of Technology (MIT) Energy Initiative. *Seminar*. MA, USA.

2023 International Energy Workshop (IEW). *Presentation*. Colorado School of Mines, CO, USA.

Skills

OS: Linux, Ubuntu, macOS, Windows, Raspbian.

Programming Languages: Python, R, Matlab, C/C++, LabVIEW, Simulink, HTML, Pascal, Bash.

Scientific Computing (HPC): CPU, GPU, and MPI parallel computing.

Prototyping: Arduino and Raspberry Pi, and 3D printing (CAD).

Libraries: TensorFlow, Keras, PyTorch, OpenCV, Scipy, Scikit-Learn.

Version Control: Git, Overleaf.

Writing: L^AT_EX, Microsoft Office, Google Docs.

Languages: Spanish (native), Portuguese (fluent).

Honors and Awards

UCSB CNSI Climate Innovation Postdoctoral Fellowship 2023-2025. Fellowship to develop technologies that contribute to achieving the climate goals of California. My investigation focuses on extending the joint day-ahead solar generation forecast to wind and electric demand.

UNM-Avangrid Graduate Scholarship, 2015. Scholarship for graduate studies at the ECE department of the UNM in the USA, within the grant King Felipe VI of Spain Endowed Chair of the UNM, sponsored by the Iberdrola Foundation (Avangrid).

UZ-Santander Bank Americampus Scholarship, 2011. Santander Bank grant for an exchange year at Latin-American universities. My goal was to learn sustainable architecture and apply my knowledge of renewable energy to develop self-sustainable energy systems.

Grants

UCSB CNSI Climate Innovation Found 2023-2024. Funding (\$30K) to accelerate the development of innovations that promote climate resilience. The funding is for computing resources and attending conferences.

UCSB IEE Research Seed Program 2022-2023. 1 year of funding (\$50K) to develop software for Smart Grids' control systems, a hierarchy of energy storage solutions, and market mechanisms. My investigation focuses on developing an efficient joint day-ahead solar forecast.

Institutional Service and Community Involvement

Mentor. UCSB EUREKA! Program 2022 for STEM undergraduate students, Santa Barbara, CA.

Mentor. UCSB CAMP Research School Year Internship 2022, Santa Barbara, CA.

Translator. Parent-teacher conferences for Spanish speakers at South Valley Academy (SVA), Albuquerque, NM.

Reviewer. *Applied Energy*, *Renewable Energy*, *Knowledge-Based Systems*, *Solar Energy*, and *Data in Brief*.

Professional References

Dr. Ranjit Deshmukh (rdeshmukh@ucsb.edu). Associate Professor. *Bren School of Environmental Science & Management, UCSB, CA, USA.*

Dr. Manel Martínez-Ramón (manel@unm.edu). Professor. *Department of Electrical and Computer Engineering, UNM, NM, USA.*

Dr. Michael Ludkovski (mludkov@ucsb.edu). Professor. *Department of Statistics and Applied Probability, UCSB, CA, USA.*